# University of AJK, Muzaffarabad

## Course Title:

## Software Design and Architecture

## Course Code:

## SEC-532

## Lab#1

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# Food Delivery System

## Task 1. Candidate Conceptual Classes

The following are the candidate conceptual classes for a Food Delivery System:

1. Customer: A person who places food orders.

2. Restaurant: A place that prepares food for customers.

3. Menu: A collection of food items offered by a restaurant.

4. Menu Item: A food item or dish available for ordering.

5. Order: A transaction created when a customer places an order.

6. Order Item: Any food item selected for an order, including its quantity.

7. Cart: A container that holds selected food items before an order is placed.

8. Delivery Driver: A person responsible for delivering orders to customers.

9. Address: The location where the order must be delivered.

10. Rating: Feedback given by a customer about food or restaurant service.

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## Task 2. Three Important Classes: Attributes and Methods

### Class 1: Customer

**Attributes (State):**

- customerName

- phoneNumber

- Address

**Methods (behavior):**

- Login()
- Place Order()
- make Payment()
- give Rating()

## Class 2: Restaurant

**Attributes (State):**

- restaurantName
- restaurantID
- location
- rating

**Methods (Behavior):**

- addMenuItem()
- acceptOrder()
- rejectOrder()
- prepareFood()

## Class 3: Order

**Attributes (State):**

- orderID
- orderDate
- status
- totalAmount

**Methods (Behavior):**

- calculateTotal()
- updateStatus()
- generateBill()
- cancelOrder()

## Task 3. Abstraction Decisions

1. We ignored the internal cooking process of the restaurant, such as cooking temperature and preparation steps, because our system focuses on ordering and delivering food.

2. We kept only the relevant customer information, such as name, customer ID, phone number, and address. We ignored details like age, height, and favorite color because they are not required for food delivery.

3. We kept important order information, such as order ID, order date, status, and total amount. We ignored details like the smell of food and the color of the delivery bag because they are not necessary for managing orders.

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## Task 4. Invented Software Object

In real life, a customer directly communicate with a waiter to place an order. However, in our software, we create an Order Controller that receives order requests from customers, sends orders to restaurants, and communicates with other objects involved in the process. This controller is a software object that helps coordinate different activities and does not have a direct equivalent in the real world.